SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	Tejhash P	**Reg. No.:**	192512495
Section / Batch:	1	**Date:**	11/10/2026

Code of Conduct

192512495

adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejhash.P

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**

8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Annexure A – Error Analysis Task

Answers are written in blue inside the tables.

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"judges … intelligent if its internal reasoning always follows the laws of logic"; "standard example of the thinking rationally approach"
S2	Incorrect	"AI winters … occurred in the 1950s, before the 1956 Dartmouth workshop"; "That workshop then revived the field" (coining the term is correct)
S3	Incorrect	"always achieves the best possible outcome because it knows the actual result of every action in advance"; "the taxi … was not rational"
S4	Incorrect (two errors)	Performance measure given as "steering, accelerator, brake, horn"; Actuators given as "safe, fast, legal and comfortable trip"
S5	Incorrect (three errors)	"partially observable"; "stochastic"; "episodic" (static and discrete are correct)
S6	Correct	None

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)
S1	Confuses behaviour-based tests with reasoning-based ones. The Turing Test judges only external behaviour (can an interrogator tell the machine from a human?), so it is **acting humanly**. "Thinking rationally" is the laws-of-thought approach (Aristotle, logic).
S2	Wrong chronology. Dartmouth (1956) is the birth of AI, so the AI winters (funding drops after over-promising) could only come later: mid-1970s and late 1980s to early 1990s.
S3	Confuses rationality with omniscience. A rational agent maximises expected performance given its percept sequence and knowledge; it cannot know actual outcomes in advance. Judging the taxi by an unforeseeable result repeats the mistake.
S4 – Performance measure	PEAS roles swapped. The performance measure is the criterion of success (a safe, fast, legal, comfortable trip), not the physical devices.
S4 – Actuators	Actuators are the means of acting (steering, accelerator, brake, horn), not a measure of success.
S5 – partially observable	The whole board is always visible, so chess is fully observable.
S5 – stochastic	No chance element: every move has a certain outcome, so chess is deterministic.
S5 – episodic	Each move affects later positions and the result, so chess is sequential.

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine intelligent if a human interrogator cannot reliably distinguish its responses from a human's. It is the standard example of the **acting humanly** approach.	It tests external behaviour only, not the form of internal reasoning.

Statement	Corrected statement	Justification
S2	The AI winters occurred later (mid-1970s and late 1980s to early 1990s). The 1956 Dartmouth workshop coined the term “artificial intelligence” and launched the field.	Dartmouth marks the birth of AI, so winters follow it.
S3	A rational agent selects the action expected to maximise its performance measure given its percept sequence and built-in knowledge. It is not omniscient, so the taxi’s choice was rational.	Rationality concerns expected, not actual, outcomes; the landslide was unforeseeable.
S4	Performance measure: safe, fast, legal, comfortable trip. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	Each PEAS component restored to its definition.
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete (and multi-agent).	Full board visible, no chance, moves have consequences, nothing changes while the player thinks.
S6	No change	The internal state tracks how the world evolves and what actions do, so the agent copes with partial observability.

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi’s choice was rational.

The four factors are: (1) the performance measure: a safe, fast trip; (2) the agent’s prior knowledge of the road network; (3) the actions available: it could choose among routes; (4) the percept sequence to date: every traffic report showed the highway clear. Given these, the highway maximised expected performance, so the choice was **rational**. The landslide could not have been predicted, and rationality does not require omniscience.

(ii) Name the properties of the taxi’s task environment that make perfect foresight impossible.

The environment is **partially observable, stochastic and dynamic** (also sequential, continuous and multi-agent). Hidden and random events such as landslides make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

A simple reflex taxi sees only the current camera frame, so it cannot tell whether a car is in its blind spot and may change lanes into it. A model-based reflex taxi keeps an **internal state**: recently seen nearby vehicles with their positions and speeds, knowledge of how the world evolves, and the effects of its own actions.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the Missionaries and Cannibals problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the fewest crossings.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	The boat's side is missing, yet it decides which crossings are possible.
Initial state	Incorrect (carried)	(3, 3) does not say the boat is on the left bank. Counted once with the state representation.
Actions	Correct	1M, 2M, 1C, 2C, 1M+1C are all the loads of one or two people; the boat is never empty.
Goal test	Incorrect	M = C is true at the start (3,3) and at (1,1), (2,2); it does not mean everyone is on the right bank.
Legality check	Incorrect	Checks only the left bank. The right bank must also have no outnumbered missionaries.
Path cost	Incorrect	Counts people carried, but the objective is the fewest crossings.
Note N1	Incorrect	Wrong phase order. Goal formulation must come first, then problem formulation, then search, then execution.
Note N2	Incorrect	16 and 10 count only (M, C) pairs and ignore the boat. With the boat the state space has 32 states, of which 20 are legal. (The student's own left-only check would also accept 13 pairs, not 10.)

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause
State representation and initial state	A state must hold all information needed to apply actions and test the goal. Leaving out the boat side violates this completeness principle.
Goal test	The goal test must describe exactly the goal condition. M = C holds at non-goal states, so the agent would stop immediately.
Legality check	The safety constraint applies on **both** banks; the student modelled only one.
Path cost	Path cost must encode the stated objective (fewest crossings), so each crossing costs 1 whatever the number of passengers.
Note N1	The problem-solving agent first formulates the goal, then the problem, then searches, then executes the plan. Search needs a defined goal and problem first.
Note N2	The state space was counted from an incomplete state description (no boat), so the counts are wrong.

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): M, C = missionaries and cannibals on the left bank; B ∈ {L, R} = boat side	Complete description of the situation
Initial state	(3, 3, L)	The boat starts on the left
Goal test	State = (0, 0, R)	Everyone, boat included, is on the right
Legality check	Left bank: M = 0 or C ≤ M. Right bank: 3−M = 0 or 3−C ≤ 3−M	Both banks must be safe
Path cost	1 per crossing	Matches "fewest crossings"
Note N1	Goal formulation → problem formulation → search → execution	Standard agent cycle
Note N2	4 × 4 × 2 = 32 states, of which 20 are legal (10 legal (M, C) pairs × 2 boat sides)	The boat side doubles the count

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal?	Reason
1M	(2, 3, R)	No	Left bank: 2 M outnumbered by 3 C
2M	(1, 3, R)	No	Left bank: 1 M outnumbered by 3 C
1C	(3, 2, R)	Yes	Left: 3 M ≥ 2 C; right: no missionaries
2C	(3, 1, R)	Yes	Left: 3 M ≥ 1 C; right: no missionaries
1M + 1C	(2, 2, R)	Yes	Left: 2 M = 2 C; right: 1 M = 1 C

(ii) Give one state that the student's legality check accepts although it is actually illegal.

The state (2, 1, ·): the student's check accepts it (left bank 2 M ≥ 1 C), but the right bank would hold 1 M and 2 C, so the missionary is outnumbered and the state is illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

11 crossings (1 per crossing). One optimal solution: 1M+1C →, 1M ←, 2C →, 1C ←, 2M →, 1M+1C ←, 2M →, 1C ←, 2C →, 1C ←, 2C →.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. Conventions: graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. BFS: do not add a node that is already explored or already in the queue. DFS: use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. UCS: write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

Neighbours (alphabetical): S: C, D | A: B, C | B: A, D, G | C: A, S | D: B, E, G, S | E: D, G | G: B, D, E

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Queue shown as [B, E, G, A]. A was already in the queue from iteration 2, so FIFO order is [A, B, E, G]. This carries into iterations 4–6 and the count of 6 removed nodes (correct: 7).

No.	Where (trace / iteration / remark)	What is wrong
2	DFS, iteration 1	Stack [D, C] pops D first. The alphabetically first child C must be on top: [C, D].
3	DFS, iteration 2	Children of D pushed as [G, E, B, C], reversed. Correct order with B on top: [B, E, G, C] (S is explored).
4	UCS, iteration 4	After expanding E (g = 5), G via E costs 8 < 9, so G:9(D) must be replaced by G:8(E). Carried into iterations 5–7.
5	Remark R1	An explored set is needed in BFS graph search.
6	Remark R2	Equal results do not prove optimality or correctness; cost 9 is not optimal (8 exists).
7	Remark R3	Fewest nodes removed does not make a strategy best.

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	BFS uses a FIFO queue: nodes already queued keep their place and new nodes join at the back. The student put A at the end.
2	A stack is LIFO, so to pop the alphabetically first child first it must be pushed last and end on top.
3	Same push-order rule violated again: children were pushed so that G ended on top instead of B.
4	UCS must replace a frontier entry whenever a cheaper path to that node is found. Missing the replacement left a suboptimal cost of 9.
5	In an undirected graph with cycles, nodes are reachable repeatedly (e.g. S from C). BFS needs the explored set to avoid re-adding them; the student's own trace also keeps one.
6	Only UCS guarantees optimality. Agreement on the same path is not evidence of optimality, and the traces contained errors.
7	Node count measures search effort, not solution quality. DFS is neither optimal nor complete in general.

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D] (D is already on the stack, so it is not pushed again)
5	G (goal)	–

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C) (G replaced: 9 → 8)
5	B	7	G:8(E), A:9(C) (A via B costs 9, not cheaper, so kept)
6	G (goal)	8	–

Justification of the corrections

Each trace now follows its data-structure rule: a FIFO queue for BFS (already-queued A keeps its place ahead of B, E, G), a LIFO stack for DFS with the alphabetically first child pushed last so it is popped first, and a cost-ordered queue for UCS in which G's entry is replaced by the cheaper path via E.

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S – D – G	3 + 6 = 9	7	No
DFS	S – C – A – B – G	3 + 6 + 2 + 3 = 14	5	No
UCS	S – D – E – G	3 + 2 + 3 = 8	6	**Yes**

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

In corrected BFS iteration 3, D is dequeued and its neighbour S is already explored. Without an explored set, S would be added to the queue again ([A, B, E, G, S]) and later re-expanded, re-adding C and D and causing loops and repeated work. The explored set (with the “already in queue” check) prevents this.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3	Suck on S6 (R, C, D) gives S8, so the belief state is {S8}, not {S6, S8}.
2	Task B, step 0 (percept update)	{S1, S2, S3, S6} includes S2 and S6, where the agent is in the Right square. Only {S1, S3} fit [Left, Dirty]. The predicted set in step 1 is carried forward from this.
3	Task B, step 2 (predicted)	Right takes S7 (L, C, C) to S8, so the set is {S6, S8}, not {S6, S7}.
4	Conclusion C1	Sensorless problems can be solved; the Task A plan ends in a goal state.
5	Conclusion C2	{S6} is not a goal: the right square is dirty.
6	Conclusion C3	There are $2^8 = 256$ belief states, not 8.
7	–	No further error.

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	The prediction step must apply the action to every state in the belief state. The student left S6 unchanged instead of cleaning it.
2	The percept update must keep only the states consistent with the whole percept (location and status of the current square).
3	Same prediction rule misapplied: Right moves the agent from the Left to the Right square, so S7 becomes S8.
4	Coercion: actions alone can shrink a belief state, and a plan is a solution if every state it ends in is a goal. A belief state with several states can still be a solution.
5	Knowing the state is not the same as reaching the goal. The goal test needs every state in the belief state to have both squares clean.
6	A belief state is any subset of the physical states, so n states give 2^n belief states, not n.

No.	Root cause (concept or rule misapplied)
7	–

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	**{S8}**

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	**{S1, S3}**
1	Suck	**{S5, S7}**	[Left, Clean]	{S5, S7}
2	Right	**{S6, S8}**	[Right, Dirty]	{S6}

(iii) Corrected conclusions C1 and C2, with justification.

C1: The plan [Suck, Right, Suck] ends in {S8}, a goal state, so a sensorless problem can be solved with certainty by coercion, even though the agent never learns its location or the dirt status.

C2: The belief state is the singleton {S6}, so the agent knows its location and dirt status. But S6 = (R, C, D) has a dirty right square, so the goal has **not** been reached. One more Suck gives S8.

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

2^8 = **256** belief states (255 non-empty). Belief-state search is harder because the space grows exponentially with the number of physical states, and each node is a whole set that must be stored, compared and updated, instead of a single state.

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}

Solution? Why?

No. S4 = (R, D, C) still has a dirty left square, so not every state in the final belief state is a goal.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____